



St. Jude Children's  
Research Hospital

# AI-Assisted Clinical Data Interface

*From Research Questions to Research-Ready Data*

Project Brief – St. Jude Biohackathon 2026  
Team KIDS26-Team2

## Abstract

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Large clinical research resources such as the **Childhood Cancer Survivor Study (CCSS)** and the **St. Jude Lifetime Cohort (SJLIFE)** hold rich longitudinal data, but using them well requires deep, dataset-specific knowledge that many investigators do not have. This project builds an AI-assisted interface that lets a clinician, investigator, analyst, or biostatistician start from a scientific question and not a database schema, and get back the cohort, variables, and evidence needed to answer it. Using **MIMIC-IV** as a proof-of-concept environment, the system combines **text-to-SQL** over structured data with **retrieval-augmented generation (RAG)** over clinical notes, chaining structured and unstructured evidence into grounded, traceable answers. The long-term goal is to adapt this framework to **CCSS** and **SJLIFE** so that St. Jude's own research resources become easier to navigate across multidisciplinary teams.

## Background & Problem Statement

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Investigators working with **CCSS**, **SJLIFE**, and comparable datasets routinely face the same barrier before any analysis can begin: they must first learn what variables exist, where they are stored, whether a given scientific question is even answerable with the available data, and how to retrieve the relevant structured and narrative fields. This dataset-fluency requirement slows down research, concentrates institutional knowledge in a small number of people, and limits how effectively multidisciplinary teams (clinicians, biostatisticians, analysts) can collaborate on a shared cohort.

## Objective

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Develop an AI-assisted interface that allows clinicians, investigators, analysts, and biostatisticians to begin their analysis with a scientific or clinical question, rather than with database schemas and table names, and returns a defensible, evidence-grounded path from that question to research-ready data.

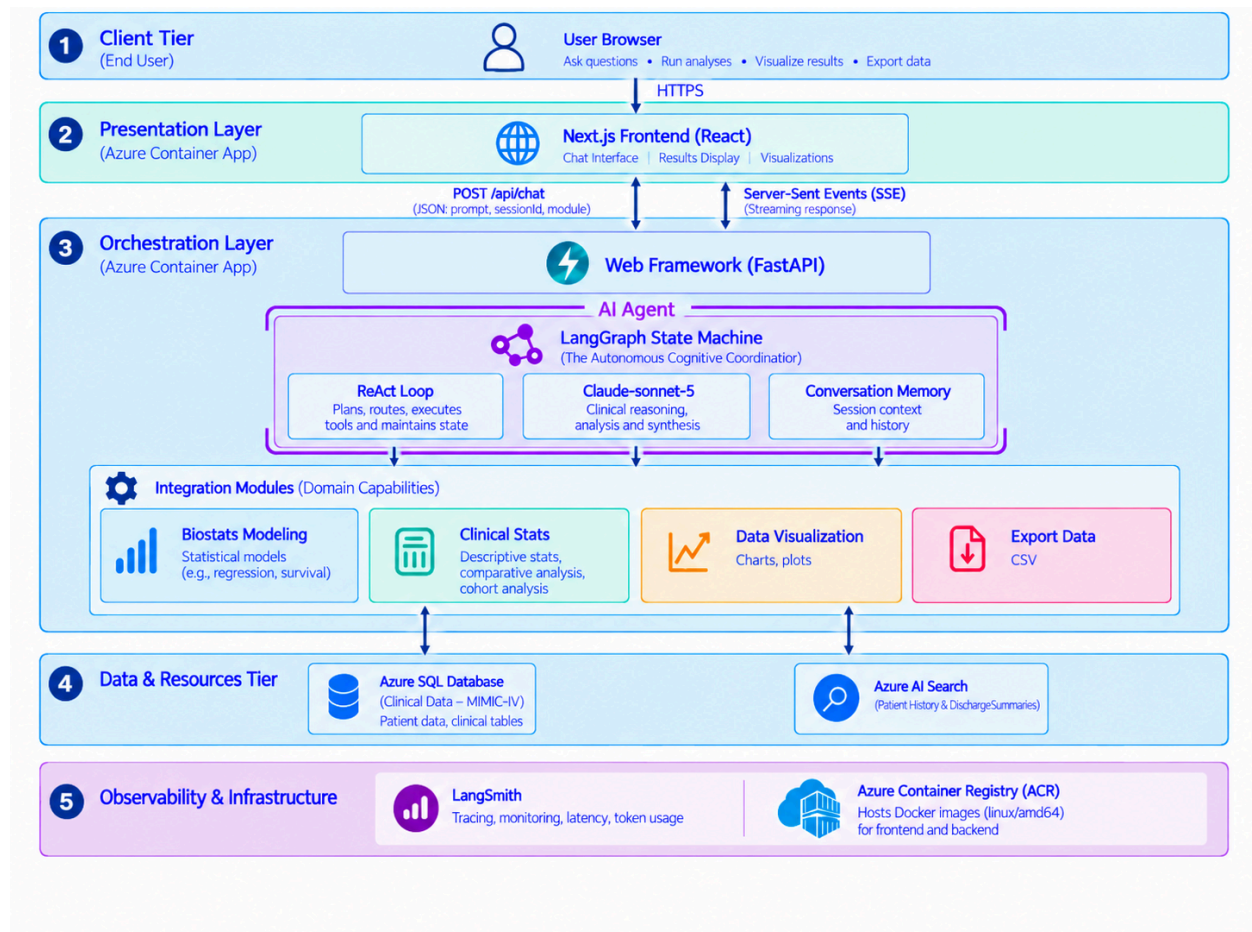
## Technical Approach & System Architecture

The proof-of-concept uses **MIMIC-IV** as a stand-in for CCSS/SJLIFE-scale clinical data. The system combines four capabilities:

- **Text-to-SQL:** for structured clinical data, so a question can be translated into a query over labs, encounters, diagnoses, and outcomes without the user writing SQL.
- **Retrieval-augmented generation (RAG):** over clinical notes, so relevant narrative evidence (e.g., radiology or discharge notes) can be surfaced alongside structured results.
- **Cohort identification:** chaining structured and unstructured data together to define and refine a patient cohort relevant to the question.
- **Evidence-grounded responses:** so every answer is traceable back to the specific rows, tables, or notes that support it, rather than being a free-text summary.

### Architecture Overview

The system is organized into a web front end, an orchestration layer, a set of specialized clinical tools, and a data layer, as shown below.



- **Frontend (Next.js UI):** the interface where a user poses a clinical or research question in plain language.
- **FastAPI & Agent: Azure Gateway:** handles request routing, auth, and orchestration entry into the agent layer.
- **FastAPI & Agent: LangGraph & Claude:** the reasoning core. A **LangGraph** agent, powered by **Claude**, plans how to answer the question and delegates to the tools below.
- **Clinical Tools:** SQL & Notes Engine (structured queries and note retrieval), Stats & Analytics, Biostats Modeler, and a Viz & CSV Exporter — each a focused capability the agent can call.
- **Data Layer:** the **MIMIC-IV** database backs the SQL/Notes, Stats, and Biostats tools; a static file store backs exported visualizations and CSVs.

## Example Use Case

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A user asks: “Are early radiographic findings in hospitalized pneumonia patients associated with ICU transfer or in-hospital mortality?” The system identifies the appropriate cohort, surfaces the available demographic and clinical variables, pulls relevant laboratory and microbiology data, retrieves ICU outcome fields, links relevant narrative notes, and reports which data sources can and cannot support the question, which gives the investigator a concrete, checkable starting point rather than a blank schema browser.

## Innovation

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The key innovation is that rather than the chatbot only answering a query, the system helps users determine which research questions are actually supportable by the available data and how the dataset should be used to address them, surfacing gaps and constraints up front instead of after a failed analysis. Grounding every answer in specific structured and narrative evidence, and chaining structured and unstructured retrieval in a single reasoning loop, distinguishes this from generic natural-language-to-SQL tools.

## Feasibility, Completeness & Evaluation

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The proof-of-concept architecture is demonstrated end-to-end on **MIMIC-IV**, combining structured SQL retrieval with **RAG** over clinical text through a working **LangGraph agent** (with dedicated **execute\_sql\_query** and **search\_discharge\_notes** tools). The dataset was deliberately migrated from **MIMIC-III** to **MIMIC-IV** to work against a more current, better-documented schema before extending to CCSS/SJLIFE-scale data.

To validate accuracy and reliability rather than relying on ad hoc spot checks, the team is building an automated evaluation harness that:

- Benchmarks text-to-SQL execution accuracy and abstention behavior using the **EHRSQL** benchmark methodology.
- Applies **sql-eval**'s pluggable-runner pattern so the same benchmark suite can run against the live agent, not just the underlying model.
- Extends coverage to **RAG** retrieval and generation quality, so note-grounded answers are checked as rigorously as SQL answers.
- Evaluates the interface/agent end-to-end, not only the underlying text-to-SQL model in isolation, so results reflect what a user actually experiences.

This gives the project a concrete, reproducible path to demonstrate correctness and a realistic plan for continued development beyond the hackathon.

## Relevance to the St. Jude Mission

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The long-term goal is to adapt this framework to St. Jude's own complex research resources, **CCSS** and **SJLIFE**, making them easier to navigate and use across multidisciplinary teams of clinicians, biostatisticians, and analysts. By lowering the dataset-fluency barrier, the system can help more of St. Jude's research staff independently determine whether a scientific question is answerable and how to pursue it, accelerating pediatric cancer survivorship research without requiring every investigator to become a database expert.

## Current Status

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- Proof-of-concept architecture demonstrated on **MIMIC-IV** with structured SQL retrieval and **RAG** over clinical text.
- Working **LangGraph agent** with SQL execution and discharge-note search tools, orchestrated behind a FastAPI/Azure gateway.
- Migration from **MIMIC-III** to **MIMIC-IV** completed to align with a more current schema.
- Automated evaluation harness (**EHRSQL**- and **sql-eval**-based) in development to benchmark the agent end-to-end.

## Acknowledgement

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We thank the St. Jude Biohackathon organizers and mentors for their guidance and feedback throughout this project, and acknowledge the use of the **MIMIC-IV** database (PhysioNet) as the proof-of-concept clinical dataset underlying this work.

## Future Work

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- Complete and run the automated evaluation harness across execution accuracy, abstention, and **RAG** retrieval/generation metrics.
- Expand the **Stats & Analytics** and **Biostats Modeler** tools to support more complex cohort and outcomes questions.
- Pilot the framework against a **CCSS** or **SJLIFE**-shaped schema to validate the adaptation path beyond **MIMIC-IV**.
- Harden evidence traceability (source citations for every structured and narrative claim) ahead of any clinical research use.

## Conclusion

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This project demonstrates that an AI-assisted interface can meaningfully lower the barrier between a clinical research question and the data needed to answer it. The **MIMIC-IV** proof-of-concept validates the core architecture: **text-to-SQL**, **RAG** over notes, and evidence-grounded responses, and lays a concrete foundation

for adapting this framework to St. Jude's own **CCSS** and **SJLIFE** resources. With the evaluation harness and expanded tooling underway, the team is well positioned to carry this work forward beyond the hackathon.